

- It makes it easy to visualize trades on charts

For beginners looking to move beyond basic backtesting, Backtrader offers an excellent balance of power and usability.

Zipline: Institutional-Grade Backtesting

Zipline, originally developed by Quantopian (now part of Robinhood), is an institutional-grade backtesting framework used by professional quants. It was designed specifically for rigorously testing algorithmic trading strategies at scale.

Here's a simplified example of our strategy in Zipline:

```
1. from zipline.api import order_target, record, symbol
2. from zipline.finance import commission, slippage
3. import pandas as pd
4.
5. def initialize(context):
6.     # Define the security we want to trade
7.     context.asset = symbol('AAPL')
8.
9.     # Set parameters
10.    context.short_period = 20
11.    context.long_period = 50
12.
13.    # Set realistic trading costs
14.    context.set_commission(commission.PerShare(cost=0.001, min_trade_cost=1.0))
15.    context.set_slippage(slippage.VolumeShareSlippage(volume_limit=0.025,
price_impact=0.1))
16.
17. def handle_data(context, data):
18.     # Get historical pricing data
19.     short_history = data.history(context.asset, 'close', context.short_period, '1d')
20.     long_history = data.history(context.asset, 'close', context.long_period, '1d')
21.
22.     # Calculate moving averages
23.     short_sma = short_history.mean()
24.     long_sma = long_history.mean()
25.
```